

بِسْمِ اللَّهِ الرَّحْمَنِ الرَّحِيمِ

GRADUATION PROJECT PROPOSAL

**A Digital Twin-Based Framework for Urban Infrastructure Resiliency and
Subsurface Utility Management in Post-Conflict Khartoum Using Machine
Learning and BIM-GIS Integration**

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Chapter 1

1.1 Introduction

Post-conflict cities often face serious difficulties in understanding the condition of their buildings and infrastructure and in deciding where recovery efforts should begin. In Khartoum, the conflict has affected important urban areas containing administrative, commercial, institutional, transportation, and other essential infrastructure. Changes to these assets can affect access, the provision of services, and the ability of the city to recover. At the same time, the availability of recent, accurate, and well-organized spatial data is limited, making it difficult to obtain a clear picture of the current urban environment.

Ground-based surveying and field inspection are important for obtaining accurate information about physical conditions. However, carrying out field surveys across affected areas may not always be practical because of safety concerns, restricted access, damaged infrastructure, time, and limited resources. Geomatics Engineering provides useful tools that can complement field-based methods through the use of Geographic Information Systems (GIS), Remote Sensing (RS), spatial databases, and geospatial analysis. Together, these technologies can help collect, organize, analyze, and visualize information about urban areas while reducing the need for extensive field observations.

Multi-temporal satellite imagery can be used to compare urban conditions before and after the conflict and to identify changes that are visible from space. When combined with Machine Learning, these data can also support the extraction and classification of urban features and the identification of observable patterns of building damage. Building footprints, road networks, and other available infrastructure information can be integrated with these results to provide a more complete spatial view of the affected area. Such information can be particularly useful where detailed ground observations are difficult to obtain.

However, assessing visible changes is only one part of managing a complex urban environment. Infrastructure information also needs to be organized so that different assets, their locations, attributes, and relationships can be understood together. The integration of GIS with Building Information Modelling (BIM) and three-dimensional modelling provides an opportunity to connect geospatial information with infrastructure-related information within a common framework. This approach can contribute to the development of a Digital Twin that represents relevant parts of the urban environment and can be updated as new information becomes available.

Subsurface utilities are also an important component of urban infrastructure. Water supply, drainage, electricity, telecommunications, and other underground services play a major role in the functioning and recovery of a city. In a post-conflict environment, information about these networks may be incomplete, outdated, fragmented, or difficult to integrate with above-ground infrastructure data. A Digital Twin-based approach can provide a way to organize available utility information, relate it to other infrastructure assets, identify gaps in the existing data, and support better infrastructure management and resilience planning.

Based on these considerations, this research proposes a Digital Twin-based framework for urban infrastructure resiliency and subsurface utility management in post-conflict Khartoum. The framework will integrate GIS, Remote Sensing, Machine Learning, spatial databases, and BIM-GIS concepts to bring together information about buildings, roads, selected infrastructure, and available subsurface utility data. The Khartoum Central Business District (CBD) will be used as the primary study area. The research will examine how these technologies can work together to support observable damage assessment, infrastructure resilience, utility management, and recovery planning, while establishing a practical geospatial foundation for future development of a more comprehensive Digital Twin.

1.2 Problem Statement

The recent armed conflict in Khartoum has caused substantial disruption to buildings, commercial and administrative facilities, transportation networks, and other urban infrastructure. The effects are not limited to individual structures, since damage or disruption to one infrastructure component can influence the operation and accessibility of other connected assets. This creates a particular challenge in the Central Business District (CBD), where buildings, roads, public facilities, and essential services are concentrated within a relatively small and highly connected urban area.

One of the main difficulties in the post-conflict environment is the lack of current, accurate, and consistently organized spatial information. Available datasets may be incomplete, outdated, or stored in different formats and sources. Information describing changes to buildings and infrastructure may also be scattered across different datasets, making it difficult to compare conditions, identify affected areas, and maintain a consistent spatial record. The situation is more challenging for subsurface utilities, since information about underground water, drainage, electricity, telecommunications, and other networks may be incomplete or difficult to access and integrate with above-ground infrastructure data.

Field surveying and physical inspection can provide important information about infrastructure conditions, but conducting these activities across affected areas may be restricted by safety concerns, limited access, damaged roads and buildings, time, and available resources. In addition, manually collecting, digitizing, and updating large amounts of urban spatial data can require considerable time and effort. These challenges highlight the need for a more organized approach that can make better use of remotely sensed data and existing spatial datasets.

Remote Sensing and Machine Learning provide opportunities to identify observable changes in buildings and selected infrastructure using multi-temporal imagery. GIS can then be used to organize these results and relate them to other spatial information such as building footprints, roads, and infrastructure networks. However, damage assessment on its own does not provide a complete solution for urban infrastructure management. There is a need to move beyond mapping individual changes and develop a framework in which different infrastructure datasets can be brought together and used to support resilience and recovery planning.

Another important issue is the limited integration between above-ground infrastructure information and subsurface utility data. Without a common spatial framework, it can be difficult to understand how these assets relate to one another, update their information, or identify areas where important data are missing. BIM-GIS integration and Digital Twin concepts provide a potential approach for bringing geometric, spatial, and infrastructure information together in a more connected environment.

Therefore, the main problem addressed by this research is the lack of an integrated and practical geospatial framework for assessing, organizing, and managing information related to post-conflict urban infrastructure in Khartoum. The absence of such a framework limits the effective use of multi-temporal imagery, Machine Learning, GIS databases, and available infrastructure information for understanding urban changes and supporting recovery.

This research will address this problem by developing a Digital Twin-based framework that integrates Remote Sensing, GIS, Machine Learning, spatial databases, and BIM-GIS integration. The Khartoum CBD, covering approximately 1.62 km², will serve as the primary study area, with WGS 84 / UTM Zone 36N (EPSG: 32636) used as the main coordinate reference system. The proposed framework will focus on observable urban changes, infrastructure resilience, and the management and integration of available subsurface utility information, while providing a structured basis for future development of a more comprehensive urban Digital Twin.

1.3.1 Research Aim

The main aim of this research is to develop a Digital Twin-based Geomatics framework for assessing and managing urban infrastructure in post-conflict Khartoum. The proposed framework will bring together Remote Sensing, Geographic Information Systems (GIS), Machine Learning, spatial databases, and BIM-GIS integration to assess observable urban damage, examine infrastructure resilience, and organize available information on subsurface utilities. The Khartoum Central Business District (CBD) will be used as a case study to

demonstrate how these technologies can be brought together within a unified spatial environment to support infrastructure assessment, recovery planning, and the future development of an urban Digital Twin.

1.3.2 Research Objectives

To achieve the main aim of the research, the following objectives will be addressed:

1. To define and establish the study area:

To define the Khartoum CBD study area, covering approximately 1.62 km², and establish a consistent spatial reference using WGS 84 / UTM Zone 36N (EPSG:32636).

2. To collect and prepare urban infrastructure data:

To collect, prepare, and standardize available spatial data for buildings, roads, and selected infrastructure from sources such as OpenStreetMap, WFS/WCS services, and suitable satellite imagery.

3. To assess observable urban changes and damage:

To use multi-temporal Remote Sensing and GIS techniques to identify and map observable changes in buildings and selected infrastructure between available pre-conflict and post-conflict datasets.

4. To develop a Machine Learning component for urban feature and damage assessment:

To evaluate suitable supervised Machine Learning methods, such as Random Forest or Support Vector Machine (SVM), for extracting and classifying selected urban features and, where sufficient reference data are available, classifying observable levels of building damage.

5. To develop an integrated spatial database:

To design and populate a structured GIS spatial database that links building, road, infrastructure, observable damage, and available subsurface utility information within a common spatial framework.

6. To integrate above-ground and subsurface infrastructure information:

To organize and spatially relate available subsurface utility data, including water, drainage, electricity, telecommunications, or other relevant networks, with above-ground urban infrastructure while identifying important data gaps and limitations.

7. To investigate BIM-GIS integration for infrastructure representation:

To prepare selected infrastructure datasets for integration between GIS and BIM environments and demonstrate how spatial and infrastructure attributes can be connected within a common information framework.

8. To develop a Digital Twin-based framework for infrastructure resilience:

To establish a conceptual and practical Digital Twin framework that connects geospatial, infrastructure, damage assessment, and available utility information to support urban infrastructure resilience, management, and recovery planning.

1.4 Research Questions

Based on the research aim and objectives, this study seeks to answer the following questions:

1. How can the Khartoum Central Business District (CBD) be spatially defined and established as a suitable study area for post-conflict urban infrastructure assessment?
2. What urban infrastructure and spatial datasets can be collected and prepared from available sources to support the assessment of the Khartoum CBD?
3. How can multi-temporal Remote Sensing and GIS techniques be used to identify and map observable changes and damage to buildings and selected infrastructure between pre-conflict and post-conflict conditions?
4. To what extent can Machine Learning methods, such as Random Forest or Support Vector Machine, support the extraction of urban features and classification of observable building changes when suitable reference data are available?
5. How can building, road, infrastructure, damage assessment, and available subsurface utility information be organized within an integrated spatial database?

6. How can available subsurface utility information, including water supply, drainage, electricity, and telecommunications, be spatially related to above-ground infrastructure, and what data gaps may affect its use?

7. How can BIM and GIS environments be integrated to improve the representation and management of selected urban infrastructure information?

8. How can the integration of geospatial, infrastructure, damage assessment, and available utility information contribute to a Digital Twin-based framework for urban infrastructure resilience and post-conflict recovery planning in Khartoum?

1.5 Significance of the Study

This research is significant because it brings together several Geomatics and digital technologies to address a practical problem in post-conflict urban infrastructure assessment in Khartoum. In such a setting, infrastructure information may be incomplete, outdated, or available in different formats and from different sources. Bringing these datasets together in a common spatial framework can provide a clearer basis for understanding the current condition of the urban environment and supporting future recovery planning.

From a technical perspective, the study will examine how Remote Sensing, GIS, Machine Learning, spatial databases, and BIM-GIS integration can work together as part of a Digital Twin-based approach. Rather than treating these technologies as separate tools, the research will explore how they can be connected to organize and analyze information on buildings, roads, observable damage, and available subsurface utilities.

The study is also relevant to urban infrastructure management. By bringing above-ground infrastructure together with available information on water supply, drainage, electricity, and telecommunications, the research will provide a more connected view of the infrastructure within the study area. At the same time, documenting missing or uncertain utility information will help identify limitations that need to be considered when using the resulting data for planning and decision-making.

In terms of post-conflict recovery, the research will provide a spatial basis for identifying observable changes in the built environment and organizing the infrastructure information that is available for the Khartoum CBD. This can support future assessment and recovery activities by making relevant spatial information easier to access, compare, and update.

However, the proposed approach is not intended to replace detailed engineering inspections or field surveys where such assessments are required.

The research also has academic value through its application of Digital Twin concepts to a post-conflict urban environment. It will provide an opportunity to examine the practical challenges of integrating BIM and GIS and applying Machine Learning when the quantity, quality, and consistency of available data may be limited.

Finally, the framework developed in this study can provide a starting point for future Digital Twin development in Khartoum. The structured spatial datasets and integration approach established through the research could later be extended with more detailed 3D models, field observations, sensor data, real-time monitoring, and more advanced infrastructure resilience analysis as additional data become available.

To establish the spatial baseline, a bounding polygon was configured and integrated with clipped vector road features (Figure 1).

CHAPTER TWO

STUDY AREA

2.1 Introduction

This chapter describes the geographical and urban context of the study area and explains why the selected part of Khartoum Central Business District (CBD) is suitable for this research. The chapter first provides a brief overview of Khartoum and its urban setting, followed by a more focused description of the CBD and the boundaries adopted for the study.

The chapter also considers the characteristics of the study area that are relevant to urban infrastructure assessment, including its road network, built environment, and the presence of different types of above-ground and subsurface infrastructure. Particular attention is given to the post-conflict context, since changes to the built environment and the availability of infrastructure information are central to the proposed Digital Twin-based framework.

For this research, the study area is defined as a specific spatial extent within the Khartoum CBD rather than as an official administrative boundary. The selected extent covers approximately 1.62 km² and provides a manageable area in which the proposed Remote Sensing, GIS, Machine Learning, spatial database, and BIM-GIS approaches can be applied and evaluated.

2.2 General Overview of Khartoum

Khartoum is the capital of Sudan and forms part of the wider metropolitan area commonly referred to as Greater Khartoum. The city occupies an important position within the country's administrative, commercial, and institutional structure. Its location at the meeting of the Blue Nile and White Nile has also played an important role in the historical development and spatial growth of the city.

The urban area of Khartoum contains a mixture of administrative, commercial, residential, institutional, and transportation-related land uses. The central part of the city contains many important roads, public and commercial facilities, and buildings that contribute to the

functioning of the wider metropolitan area. This concentration of activities makes central Khartoum particularly relevant when considering the relationship between urban development and infrastructure.

Khartoum is also part of a wider urban system that includes Khartoum North (Bahri) and Omdurman. These areas are connected through major roads and bridges and form an interconnected metropolitan environment. Consequently, infrastructure disruption in one part of the metropolitan system can have effects beyond the immediate location of the affected asset.

The availability of spatial data is an important consideration for studying Khartoum. OpenStreetMap, for example, provides openly accessible geographic information that can be used to identify and organize features such as roads, buildings, and other mapped urban elements. However, the completeness and positional accuracy of open geographic data can vary between locations and feature types. Therefore, such datasets need to be treated as useful spatial inputs rather than as a complete representation of all infrastructure in the city.

2.3 Urban Context of Khartoum

The development of Khartoum has resulted in a diverse urban structure, with differences in land use, building density, road hierarchy, and infrastructure provision across different parts of the city. The central urban zone has historically been associated with commercial and administrative activities and contains a relatively concentrated network of streets and urban facilities.

Previous studies of Khartoum's central area have identified issues related to transportation, parking, pedestrian movement, land use, and the organization of the built environment. These characteristics make the CBD an important area for spatial analysis, particularly when infrastructure condition and accessibility are considered together.

The urban structure is also important from a Geomatics perspective. Buildings, roads, and infrastructure networks can be represented as spatial features and linked through a common coordinate reference system. This makes the CBD a suitable environment for investigating how different spatial datasets can be integrated into a single framework.

For this research, the urban environment is therefore considered not simply as a collection of individual buildings and roads, but as a connected spatial system. Buildings are related to

roads and surrounding infrastructure, while subsurface utilities may support the same areas without being directly visible from conventional imagery. This distinction is particularly important when developing a framework intended to combine above-ground observations with available subsurface information.

2.4 Khartoum Central Business District (CBD)

The Khartoum Central Business District represents one of the most significant central urban areas of the city. It contains a concentration of commercial, administrative, institutional, and transportation activities and is connected to major roads serving the central part of Khartoum.

The CBD is particularly relevant to this research because its urban infrastructure is spatially interconnected and includes different types of assets within a relatively compact area. Buildings and road networks form the most visible components of this environment, while services such as water supply, drainage, electricity, and telecommunications are associated with both above-ground and subsurface infrastructure systems.

The concentration of infrastructure within the CBD also makes it suitable for testing a Digital Twin-based approach. Within a limited spatial extent, different datasets can be brought together, compared, and organized using GIS. Observable changes identified from multi-temporal imagery can then be linked to existing building and infrastructure information, while available utility information can be incorporated where suitable data exist.

It should be emphasized that the CBD boundary used in this research is a project-defined study boundary established for the purpose of spatial analysis. It should not be interpreted as an official administrative boundary of the Khartoum locality.

2.5 Location and Geographic Extent of the Study Area

The selected study area is located within the central part of Khartoum and covers approximately 1.62 km². The area was selected to capture a representative section of the CBD containing a combination of major roads, built-up areas, commercial and institutional properties, and infrastructure features relevant to the research.

For spatial processing and analysis, the study area is referenced using WGS 84 / UTM Zone 36N (EPSG:32636). This projected coordinate reference system provides metric coordinates suitable for measuring distances and areas and for carrying out GIS operations within the study area.

The approximate study boundary adopted for the research is defined by the following major urban corridors:

- ❖ North: Al-Jamhuriya Street
- ❖ East: Al-Mek Nimr / Al-Jaish Avenue
- ❖ South: the railway corridor near Souq Al-Arabi
- ❖ West: Al-Qasr Avenue

These boundaries are used to establish a consistent spatial extent for data collection, preprocessing, analysis, and visualization. The final boundary is represented as a polygon in the project geodatabase and is used to clip the different spatial datasets to a common study extent.

Table 2.1. General characteristics of the study area

Parameter	Description
Study area	Selected section of Khartoum CBD
Approximate area	1.62 km ²
Coordinate Reference SystemWGS 84	UTM Zone 36N
EPSG Code	32636
Study boundary type	Project-defined spatial boundary
Northern limit	Al-Jamhuriya Street
Eastern limit	Al-Mek Nimr / Al-Jaish Avenue
Western limit	Al-Qasr Avenue
Main spatial features	Buildings, roads, infrastructure, and available utility information

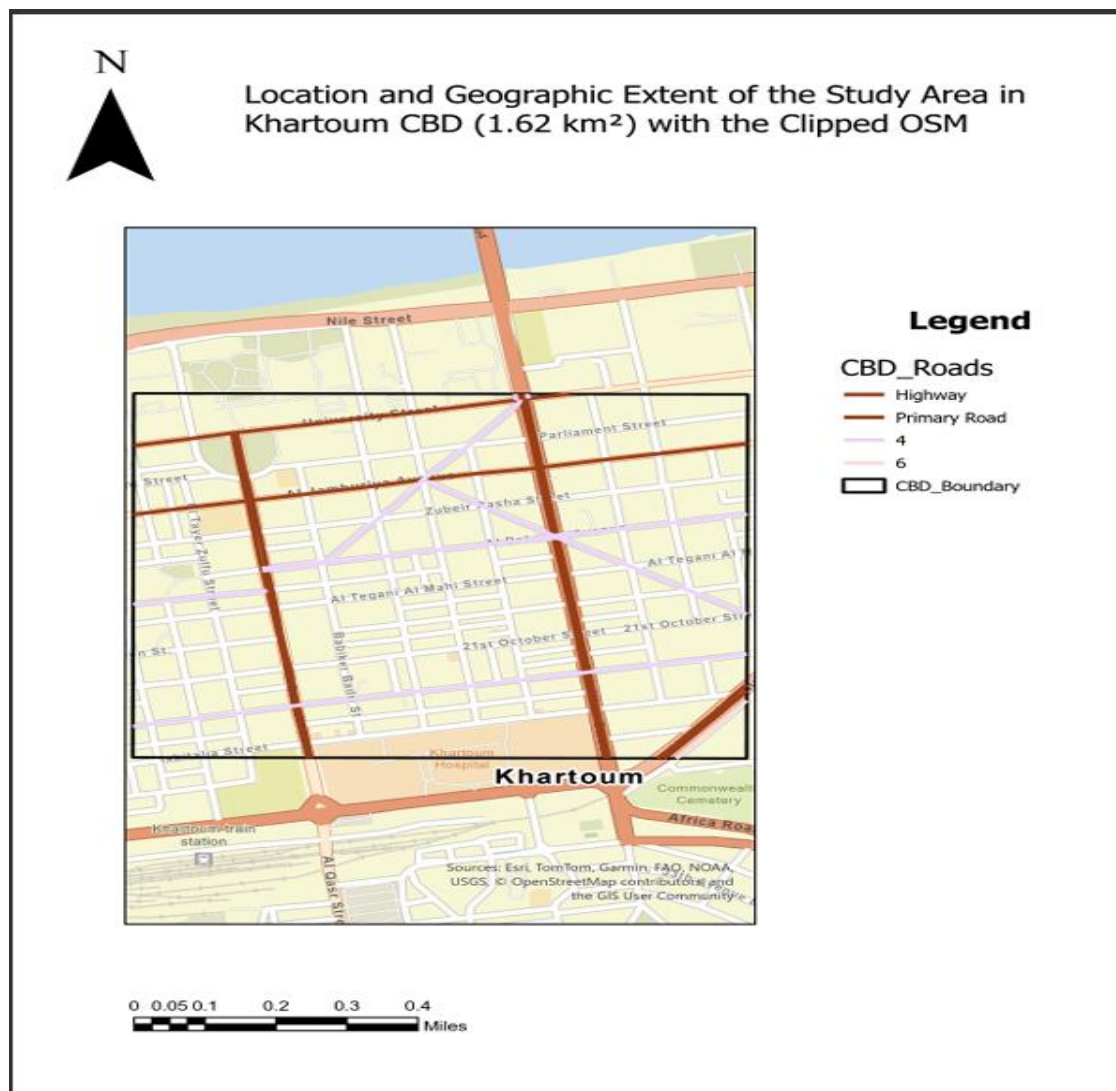


Figure 2.1 should present the location of the study area within Khartoum and show its geographic relationship to the wider urban area.



- CBD_Roads
- ▭ CBD_Boundary

0 0.03 0.05 0.11 Miles
|-----|

Figure 2.2 should provide a detailed map of the selected CBD extent, including the study boundary and the main surrounding roads.

2.6 Built Environment and Infrastructure Characteristics

The study area contains a relatively dense urban fabric in which buildings and transportation infrastructure are closely connected. Buildings vary in their function, size, footprint, and spatial arrangement, reflecting the mixture of commercial, administrative, institutional, and other urban activities found within the central area.

The road network provides an important spatial structure for the study area. Major roads form connections between different parts of central Khartoum, while smaller streets provide access to individual blocks and properties. From a GIS perspective, this network can be represented as a set of connected linear features and related to buildings and other infrastructure through their spatial relationships.

The built environment is particularly important for the damage-assessment component of this research. Building footprints provide a spatial reference against which observable changes can be examined using multi-temporal satellite imagery. Changes in building presence, footprint condition, or visible surface characteristics may provide indicators of possible damage or alteration. However, such observations represent observable changes from available spatial data and should not be interpreted as a complete structural engineering assessment.

Infrastructure in the study area can broadly be considered in two groups. The first includes infrastructure that can be represented or observed above ground, such as buildings, roads, and selected visible facilities. The second includes subsurface or partially hidden systems, such as water supply, drainage, electricity, and telecommunications networks.

This distinction is important because the second group is generally more difficult to observe directly using conventional satellite imagery. Where utility datasets are available, they can instead be spatially organized and related to the visible urban environment within the proposed GIS and Digital Twin framework.

2.7 Post-Conflict Context

The selection of Khartoum as the study location is closely related to the effects of the conflict that began in April 2023. The conflict has caused extensive disruption to urban life and infrastructure across Sudan, including damage to or disruption of roads, power, water, telecommunications, health facilities, and other essential services.

Within this context, Khartoum provides an important case for examining how spatial technologies can support the assessment of urban change. Physical access to affected areas may be difficult or restricted, while conventional infrastructure records may not always be complete or readily available. Remote Sensing therefore provides an important source of information because it allows changes in the built environment to be examined from imagery acquired at different points in time.

The post-conflict context also creates a need to bring different types of spatial information together. A satellite image can provide evidence of visible changes, but it does not by itself provide detailed information about roads, utility networks, infrastructure attributes, or historical records. GIS provides the spatial environment for organizing these different datasets, while Machine Learning can be used to assist with selected classification and feature-extraction tasks where adequate reference data are available.

The situation also highlights the importance of recognizing uncertainty. Not every visible change can automatically be attributed to conflict-related damage, and the absence of a mapped utility does not necessarily mean that the utility does not exist. For this reason, the proposed study will distinguish between observed information, derived information, and information that remains uncertain or unavailable.

2.8 Data Availability within the Study Area

The availability and quality of spatial data are important considerations in selecting and studying the CBD. The research will make use of data from several sources, depending on their availability, spatial coverage, temporal relevance, and suitability for the intended analysis.

OpenStreetMap can provide useful baseline information on roads, buildings, and selected urban features. Its open structure also makes it suitable for integration into GIS-based workflows. However, OpenStreetMap documentation indicates that the completeness of mapped features can vary, and some areas or feature types may contain limited information.

Additional spatial information may be obtained from available WFS/WCS services and suitable satellite imagery. These sources can complement vector-based datasets by providing raster or service-based information that can be used for environmental and urban analysis.

For subsurface utilities, the situation is more challenging. Information on water supply, drainage, electricity, and telecommunications may not be consistently available at the same spatial resolution or in the same format as above-ground datasets. Consequently, the research will incorporate available utility information where it can be obtained and will explicitly document gaps rather than attempting to infer unavailable networks as if they were known.

This approach is important for the Digital Twin concept adopted in the research. A useful Digital Twin framework depends not only on the availability of data, but also on knowing the reliability, source, and limitations of those data.

2.9 Suitability of the Study Area for the Research

The selected Khartoum CBD is considered suitable for this research for several related reasons.

First, the area contains a mixture of buildings, roads, and infrastructure within a relatively compact spatial extent. This provides an appropriate environment for testing the integration of multiple spatial datasets.

Second, the area provides an opportunity to examine observable changes in the built environment through multi-temporal Remote Sensing. Existing building and road information can provide a spatial reference for comparing conditions before and after the conflict.

Third, the concentration of infrastructure creates an appropriate setting for investigating relationships between above-ground and available subsurface information. Although complete utility records may not be available, the available information can still be organized and evaluated within the proposed spatial framework.

Fourth, the CBD provides a practical test case for BIM-GIS integration. Selected infrastructure features can be represented using their spatial and attribute information and prepared for integration between geospatial and BIM-based environments.

Finally, the relatively limited extent of approximately 1.62 km² makes the study manageable within the time, data, and computational constraints of a graduation research project. At the same time, the area is sufficiently complex to demonstrate the proposed Digital Twin-based framework and identify challenges that could be addressed in larger future studies.

2.10 Chapter Summary

This chapter presented the geographical and urban context of the selected study area. The research focuses on a project-defined section of the Khartoum Central Business District covering approximately 1.62 km², referenced using WGS 84 / UTM Zone 36N (EPSG:32636).

The study area contains a combination of buildings, roads, commercial and institutional activities, and infrastructure systems, making it suitable for investigating the integration of different Geomatics technologies. Its post-conflict context further provides a practical setting for examining observable urban changes and organizing available infrastructure information.

The chapter also highlighted the importance of data availability and uncertainty, particularly for subsurface utilities. Rather than assuming that all infrastructure information is available or complete, the research will identify and document existing data gaps as part of the proposed framework.

The characteristics discussed in this chapter provide the spatial foundation for the methodology presented in the following chapter, where the data sources, preprocessing procedures, Remote Sensing and GIS analysis, Machine Learning component, spatial database design, BIM-GIS integration, and Digital Twin framework will be described.

CHAPTER THREE

METHODOLOGY

3.1 Introduction

This chapter presents the methodology that will be followed to develop the proposed Digital Twin-based Geomatics framework for urban infrastructure assessment and management in post-conflict Khartoum. The methodology is designed around the integration of spatial data from different sources rather than relying on a single dataset or technology.

The research will combine Remote Sensing, Geographic Information Systems (GIS), Machine Learning, spatial database management, and BIM-GIS integration within a common spatial framework. The main purpose is to organize available information, identify observable changes in the built environment, assess selected infrastructure features, and establish the structure needed for a future urban Digital Twin.

The methodology is also designed to account for the practical limitations of working in a post-conflict environment. Data may be incomplete, available at different spatial and temporal resolutions, or missing entirely for some infrastructure systems, particularly subsurface utilities. Therefore, the study will distinguish between directly available data, information derived through analysis, and information that remains unavailable or uncertain.

The overall workflow consists of several connected stages: study-area preparation, data acquisition, data preprocessing, multi-temporal urban change assessment, Machine Learning-based classification, spatial database development, subsurface utility integration, BIM-GIS preparation, Digital Twin framework development, and validation.

3.2 Research Design

This research will follow an applied and spatially oriented research design. The study will use existing geospatial and remote sensing data to investigate observable changes in the urban environment and to develop an integrated framework for infrastructure information management.

The research will primarily use a quantitative geospatial approach, since most of the analysis will involve spatial datasets, satellite imagery, feature extraction, classification, spatial relationships, and measurable changes. At the same time, a qualitative element will be included when evaluating data availability, identifying information gaps, and assessing the practical suitability of different datasets for Digital Twin development.

The research can be summarized into five main stages:

1. Data preparation, including the collection, cleaning, standardization, and organization of available datasets.
2. Urban change and damage assessment, using multi-temporal Remote Sensing and GIS techniques.
3. Machine Learning analysis, using suitable supervised classification methods where sufficient reference data are available.
4. Infrastructure data integration, including the development of a spatial database and the organization of available subsurface utility information.
5. Digital Twin framework development, including BIM-GIS integration and the connection of the different spatial datasets within a unified framework.

The final result will be a prototype framework rather than a fully operational real-time Digital Twin. The framework will establish the spatial and data structure that could be expanded in future research when more detailed infrastructure records, field observations, sensors, or real-time data become available.

3.3 Overall Methodological Workflow

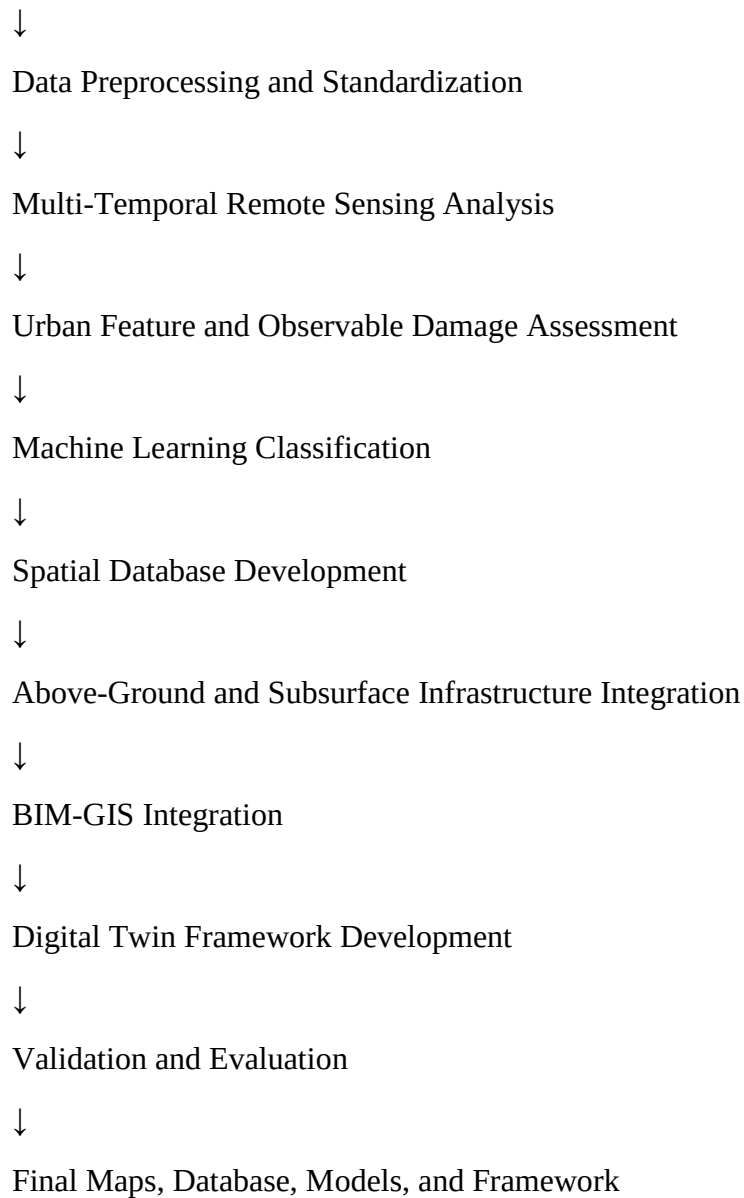
The proposed methodology follows a sequential but interconnected workflow. Each stage produces information that is used by the following stage.

The general workflow is:

Study Area Definition

↓

Data Acquisition



This workflow allows the research to move from raw and heterogeneous data toward an integrated infrastructure information environment.

Figure 3.1 should present the overall methodological workflow of the research.

3.4 Study Area Preparation

The first stage of the research will be the preparation of the study area established in Chapter 2. The selected section of the Khartoum CBD covers approximately 1.62 km² and will be represented as a polygon feature within the project geodatabase.

All spatial datasets used in the research will be transformed, where necessary, into WGS 84 / UTM Zone 36N (EPSG:32636). Using a common projected coordinate reference system will ensure consistency when calculating distances and areas and when performing spatial operations such as buffering, clipping, overlay, and proximity analysis.

The study-area polygon will also be used to clip larger datasets to the research extent. This will reduce unnecessary processing and ensure that the different datasets used throughout the study correspond to the same spatial boundary.

3.5 Data Acquisition

The research will use several types of spatial and remote sensing data. The final dataset selection will depend on availability, spatial resolution, temporal coverage, quality, and relevance to the research objectives.

3.5.1 OpenStreetMap Data

OpenStreetMap (OSM) will be used as one of the main sources of baseline vector information. Relevant features may include:

- Building footprints
- Road networks
- Major streets
- Selected points or infrastructure-related features where available

The data will be downloaded or extracted for the study area and converted into suitable GIS layers.

Because OSM is a collaboratively maintained dataset, its completeness may vary between features and locations. Therefore, OSM data will be treated as a baseline spatial source and will be evaluated before being used in subsequent analysis.

3.5.2 Satellite Imagery

Satellite imagery will be used to examine the physical characteristics and observable changes within the study area.

Suitable pre-conflict and post-conflict imagery will be identified based on:

- Acquisition date
- Spatial resolution
- Cloud coverage
- Image quality
- Coverage of the study area
- Availability of comparable imagery

Optical satellite imagery may be used to identify changes in building and surface characteristics, while radar imagery may be considered where appropriate because it can provide additional information independent of daylight and, under suitable conditions, cloud cover.

The selection of imagery will ultimately depend on what is available for the relevant dates and whether the imagery is sufficiently consistent for comparison.

3.5.3 WFS/WCS and Other Spatial Services

Where suitable datasets are available, Web Feature Service (WFS) and Web Coverage Service (WCS) sources will be used to obtain additional vector or raster information.

These services may provide supplementary environmental or spatial datasets that can support the analysis. The datasets will be assessed before integration to determine their coordinate reference system, spatial resolution, metadata, temporal relevance, and compatibility with the project database.

3.5.4 Infrastructure and Utility Data

Available infrastructure information will be collected where accessible. This may include information related to:

- Water supply
- Drainage
- Electricity
- Telecommunications
- Roads
- Selected public or critical infrastructure

The research will not assume that complete utility records exist for the study area. Where information is missing, the gap will be recorded as part of the data-quality assessment rather than filled through unsupported assumptions.

3.6 Data Preprocessing and Standardization

Before analysis, the collected datasets will undergo preprocessing to ensure that they can be used together within the same spatial environment.

The main preprocessing operations will include:

1. Checking the coordinate reference system of each dataset.
2. Reprojecting datasets to EPSG:32636 where required.
3. Removing duplicate or clearly invalid features.
4. Checking geometry validity.
5. Clipping datasets to the study-area boundary.
6. Standardizing attribute fields and naming conventions.
7. Checking spatial alignment between datasets.
8. Documenting the source and date of each dataset.
9. Recording relevant data-quality limitations.

For satellite imagery, preprocessing may include image selection, spatial subsetting, and other appropriate corrections depending on the imagery source and analysis requirements.

The purpose of this stage is not simply to prepare data for analysis, but also to establish a consistent information structure that can later be incorporated into the spatial database and Digital Twin framework.

3.7 Multi-Temporal Remote Sensing Analysis

Multi-temporal Remote Sensing will be used to compare the observable condition of the study area at different points in time.

The analysis will focus on identifying changes that can be detected from available imagery, particularly changes affecting buildings and selected infrastructure-related features.

A pre-conflict image or dataset will provide a baseline representation of the urban environment, while a suitable post-conflict image will be used to identify subsequent changes.

The analysis may include:

- Visual comparison of imagery
- Image differencing
- Spectral or backscatter-based change indicators where appropriate
- Building-level comparison
- Spatial overlay with existing building footprints
- Identification of areas requiring further classification

The results will be interpreted carefully because a detected change does not automatically indicate conflict-related damage. Changes may also result from construction, demolition, temporary land-cover changes, image differences, or other causes.

For this reason, the research will use the term observable urban change or damage where the available evidence does not allow a more specific conclusion.

3.8 Building and Urban Feature Extraction

Existing building footprints will be used as an important spatial reference for the analysis.

Where suitable imagery and reference data are available, automated or semi-automated feature extraction methods may be evaluated to identify building-related features. The purpose is to examine whether Machine Learning can assist in extracting or classifying relevant urban features within the study area.

The extracted results will be compared with available reference information. Where automated extraction does not provide sufficient reliability, existing verified spatial data will be retained as the primary reference.

This approach allows the research to evaluate Machine Learning as a practical component of the framework rather than assuming that automated extraction will always outperform conventional GIS data preparation.

3.9 Observable Damage Assessment

The damage-assessment component will focus on visible and spatially detectable changes rather than structural engineering damage assessment.

The analysis will compare available pre-conflict and post-conflict information to identify changes in building condition or presence. Depending on the quality of the available imagery and reference information, buildings may be grouped into categories such as:

- Intact or apparently unchanged
- Moderately changed

- Severely changed
- Completely destroyed or no longer observable

These categories will only be applied where sufficient evidence exists. The classification will be based on observable spatial or image characteristics and should not be interpreted as a structural safety assessment.

The resulting damage or change layer will be linked to individual building features through a unique identifier. This will allow each building to retain its original spatial and attribute information while also storing the results of the change assessment.

3.10 Machine Learning Component

Machine Learning will be used as a supporting analytical component within the proposed framework.

The study will evaluate suitable supervised classification methods, with Random Forest (RF) and Support Vector Machine (SVM) considered as candidate approaches. The final model selection will depend on the characteristics and quantity of available training and reference data.

3.10.1 Random Forest

Random Forest can be used to classify urban features based on a combination of input variables. Its ensemble structure makes it suitable for handling multiple predictors and non-linear relationships.

For this research, Random Forest may be evaluated for tasks such as:

- Urban feature classification
- Building-related classification
- Observable change classification
- Land-cover or surface-feature classification where relevant

3.10.2 Support Vector Machine

Support Vector Machine will be considered as an alternative supervised classification approach. It can be useful for classification problems where the available training dataset is relatively limited and where classes can be separated using an appropriate feature space.

The performance of SVM will be compared with other candidate approaches where sufficient reference data are available.

3.10.3 Training and Reference Data

The quality of the Machine Learning component will depend heavily on the availability of suitable reference data.

Reference samples will be prepared from reliable spatial information and image interpretation where possible. The data will be divided into training and validation subsets to reduce the risk of evaluating the model using the same information used to train it.

If sufficient reference data cannot be obtained, the Machine Learning component will be treated as an exploratory evaluation rather than presented as a fully validated operational model.

3.11 Machine Learning Evaluation

The classification results will be evaluated using appropriate accuracy measures.

A confusion matrix will be used to compare predicted classes with reference classes. Where the number and quality of reference samples allow it, the evaluation may include:

- Overall accuracy
- User's accuracy
- Producer's accuracy

- F1-score
- Class-specific performance

The evaluation will help determine whether the selected Machine Learning approach provides sufficiently reliable results for inclusion in the proposed framework.

Rather than selecting a model simply because it produces a classification map, the research will consider both the classification performance and the quality of the underlying training data.

3.12 Spatial Database Development

A central component of the methodology will be the development of an integrated spatial database.

The database will be designed to organize the different datasets used by the research and to maintain relationships between infrastructure features.

The database may contain thematic layers or feature classes representing:

- Buildings
- Roads
- Infrastructure features
- Observable damage or change
- Water supply
- Drainage
- Electricity
- Telecommunications
- Study-area boundary
- Remote sensing-derived information
- Machine Learning classification results

Each feature will contain relevant attributes and, where possible, a unique identifier.

The database structure will also include metadata describing the source, date, spatial reference, and reliability of important datasets.

This structure will make it possible to distinguish between original source information and information derived through analysis.

3.13 Above-Ground and Subsurface Infrastructure Integration

One of the distinctive components of the research is the integration of above-ground and available subsurface infrastructure information.

Above-ground features such as buildings and roads can generally be mapped using satellite imagery, OSM, and other spatial datasets. Subsurface utilities, however, are less directly observable and depend largely on existing records or other available information.

The research will therefore spatially relate available utility information to the above-ground infrastructure environment.

For example, utility datasets may be linked spatially to:

- Building footprints
- Road corridors
- Infrastructure facilities
- Administrative or functional zones

The analysis may use spatial relationships such as intersection, proximity, containment, and network connectivity where the available data support these operations.

An important output of this stage will be a utility data-gap layer or inventory identifying areas or utility types for which information is missing, incomplete, or uncertain.

This is particularly important for a Digital Twin because uncertainty in the source data needs to remain visible rather than being hidden by the final visualization.

3.14 BIM-GIS Integration

The research will investigate the integration of GIS-based infrastructure information with BIM-based representation.

The purpose is not to create detailed Building Information Models for every building in the CBD. Instead, selected infrastructure features will be prepared as representative BIM-GIS integration cases.

The process will involve:

1. Selecting suitable infrastructure features.
2. Preparing their spatial geometry and attributes.
3. Ensuring consistent coordinate reference information.
4. Organizing relevant GIS attributes.
5. Preparing selected features for representation within a BIM environment.
6. Relating BIM elements to their corresponding GIS features.

Where appropriate, building or infrastructure representations may be developed at a relatively simple level of detail. The level of detail will depend on the available source information and the requirements of the research.

The main purpose is to demonstrate the relationship between geographic location and infrastructure information rather than to produce construction-ready BIM models.

3.15 Digital Twin-Based Framework Development

The final integration stage will bring together the outputs from Remote Sensing, GIS, Machine Learning, spatial databases, infrastructure datasets, and BIM-GIS integration.

The proposed Digital Twin framework will be organized around several connected information layers:

Layer 1: Spatial Reference

This layer provides the geographic foundation of the system, including the study-area boundary, coordinate reference system, buildings, roads, and other base spatial features.

Layer 2: Infrastructure Information

This layer contains available information about infrastructure assets and their attributes.

Layer 3: Observable Change and Damage

This layer contains the results of multi-temporal Remote Sensing and Machine Learning analysis.

Layer 4: Subsurface Utility Information

This layer contains available information on water supply, drainage, electricity, and telecommunications, together with information gaps and uncertainty where applicable.

Layer 5: BIM Representation

Selected infrastructure elements will be connected to BIM-based representations to provide an additional level of infrastructure description.

Layer 6: Management and Visualization

The integrated information will be presented through GIS-based maps, tables, 3D representations, or a prototype interface where feasible.

The framework will allow different information types to be connected through common spatial identifiers and geographic relationships.

The resulting system should therefore be understood as a Digital Twin-based framework or prototype, rather than a fully operational real-time Digital Twin. Its purpose is to establish the data, spatial, and integration structure required for future development.

3.16 Data Integration and Spatial Relationships

A key principle of the proposed framework is that individual datasets should not remain isolated.

For example, a building can have:

- A geographic location
- A building footprint
- Associated attributes
- An observable change classification
- A damage-assessment category
- Nearby roads
- Associated infrastructure
- Available utility relationships
- A corresponding BIM representation where applicable

These relationships will be implemented through spatial joins, unique identifiers, relational database structures, and other suitable GIS operations.

This approach allows information from different sources to be accessed through the same spatial feature and provides a more useful representation of the urban environment than maintaining separate, disconnected datasets.

3.17 Validation and Quality Assessment

Validation will be carried out at several levels of the research.

3.17.1 Spatial Data Validation

Spatial datasets will be checked for:

- Coordinate consistency
- Geometry errors
- Duplicate features
- Missing attributes
- Spatial misalignment
- Incomplete coverage

3.17.2 Remote Sensing Validation

Observed changes will be compared against available reference imagery and spatial information. Where possible, visual interpretation and independent reference samples will be used to assess whether identified changes correspond to actual observable changes.

3.17.3 Machine Learning Validation

Machine Learning outputs will be evaluated using independent validation samples and appropriate classification metrics.

3.17.4 Database Validation

The spatial database will be checked to ensure that feature identifiers, attribute relationships, coordinate systems, and spatial relationships are consistent.

3.17.5 Framework Validation

The final framework will be evaluated according to whether it successfully connects the main data components and supports the intended infrastructure assessment and management tasks.

The validation stage will also document limitations. A result will not be considered reliable simply because it can be displayed on a map; its source, uncertainty, and validation status will also be considered.

3.18 Data Quality and Uncertainty

Data quality is expected to be one of the main challenges of the research.

Different datasets may vary in:

- Spatial accuracy
- Temporal accuracy
- Completeness
- Resolution
- Attribute quality
- Coordinate reference system
- Source reliability

These differences can affect the results of spatial analysis and Machine Learning classification.

To address this issue, each major dataset will be accompanied by basic information describing its source, date, spatial characteristics, and known limitations.

Where uncertainty cannot be eliminated, it will be explicitly documented. This is especially important for subsurface utility information, where missing records may prevent a complete representation of the infrastructure network.

The framework will therefore distinguish between:

Known information → Derived information → Uncertain information → Missing information

This distinction will help prevent unsupported assumptions from becoming part of the Digital Twin representation.

3.19 Software and Technical Environment

The research will use a combination of GIS, Remote Sensing, database, Machine Learning, and BIM-related software.

The main technical environment may include:

Component	Proposed Tools / Technologies
GIS analysis	QGIS / ArcGIS Pro
Open spatial data	OpenStreetMap
Remote Sensing	Suitable satellite imagery and GIS/Remote Sensing tools
Spatial database	GeoPackage / PostGIS / SpatiaLite
Machine Learning	Python and appropriate ML libraries
Data processing	Python
BIM	Autodesk-based BIM environment where available
Data exchange	GeoJSON, Shapefile, GeoPackage and suitable BIM formats
Coordinate system	WGS 84 / UTM Zone 36N (EPSG:32636)

Visualization	GIS maps and selected 3D/BIM representations
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The final software selection may be adjusted according to data availability, hardware requirements, software access, and compatibility between the different components.

3.20 Reproducibility and Data Management

To make the research easier to reproduce and update, datasets and processing steps will be organized systematically.

Where possible, the project will maintain:

- Clearly named datasets
- Consistent coordinate reference systems
- Documented data sources
- Standardized attribute fields
- Separate raw and processed datasets
- Reusable processing scripts
- Documented Machine Learning parameters
- Versioned outputs

Automated scripts may be developed for repetitive tasks such as data preparation, coordinate transformation, spatial clipping, attribute standardization, and database updates.

This approach will reduce unnecessary manual processing and make it easier to update the framework if new infrastructure or remote sensing data become available.

3.21 Ethical and Practical Considerations

The research will rely primarily on spatial and remotely sensed information and will not require the collection of personal information.

Care will be taken when interpreting post-conflict imagery and infrastructure conditions. The study will focus on spatial and infrastructure-related observations rather than attempting to identify individuals or sensitive personal information.

In addition, the research will avoid presenting uncertain infrastructure information as confirmed. Where the source or reliability of a dataset is unclear, this limitation will be recorded.

The results will also be presented as a research-based spatial assessment and framework rather than as a substitute for professional engineering inspections, detailed structural assessments, or official infrastructure records.

3.22 Expected Methodological Outputs

Each methodological stage is expected to produce a specific output that contributes to the final framework.

Methodological Stage	Expected Output
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Study area preparation	Standardized 1.62 km ² study boundary
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Data acquisition	Raw spatial and remote sensing datasets
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Data preprocessing	Clean and standardized GIS layers
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Remote Sensing analysis	Multi-temporal change information
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Building analysis	Building-level observable change information
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Machine Learning	Urban feature / damage classification results
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Spatial database	Integrated infrastructure geodatabase
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Utility integration	Available subsurface utility layers and data-gap information
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BIM-GIS integration	Selected infrastructure BIM-GIS representations
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Digital Twin development| Integrated Digital Twin-based framework

Validation| Accuracy and quality assessment

Final visualization| Maps, tables, spatial layers, and selected 3D representations

3.23 Chapter Summary

This chapter presented the methodology proposed for developing a Digital Twin-based Geomatics framework for post-conflict urban infrastructure assessment in Khartoum.

The methodology begins with the definition and preparation of the study area and continues with the collection and standardization of spatial, infrastructure, utility, and satellite data. Multi-temporal Remote Sensing and GIS techniques will then be used to identify observable changes in the built environment. Where suitable reference data are available, Machine Learning methods such as Random Forest and Support Vector Machine will be evaluated for urban feature and observable damage classification.

The resulting information will be organized within an integrated spatial database and spatially related to available subsurface utility information. The research will then investigate BIM-GIS integration using selected infrastructure features before bringing the different information components together within a Digital Twin-based framework.

Throughout the methodology, data quality, uncertainty, and availability will be treated as important parts of the research rather than as secondary issues. The final outcome will be a structured and validated framework that demonstrates how different Geomatics technologies can be integrated to support infrastructure assessment, resilience analysis, and post-conflict recovery planning in the Khartoum CBD.

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